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Ratan Saha, Ujjal Mandal

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Estimation of optoacoustic spectra for normal and pathological red blood cells using a modified Green's function approach

Ratan K Saha, Ujjal Mandal

Department of Applied Sciences, Indian Institute of Information Technology Allahabad, Prayagraj, 211015, India
*E-mail: ratank.saha@iiita.ac.in

Abstract

The frequency domain optoacoustic (OA) wave equation is inherently inhomogeneous. The first inhomogeneous term arises because of the OA effect (i.e., conversion of optical energy into acoustical energy). The second term appears due to sound-speed mismatch between the source and the ambient medium. The conventional Green's function method works well in absence of the second term (i.e., acoustically homogeneous source). Recently, it has been shown that a modified Green's function (MGF) approach provides faithful solution to the OA wave equation for an acoustically inhomogeneous source. Herein, we employ the MGF technique for accurate estimation of the OA spectra for normal and pathological red blood cells (RBCs). The shapes in 2D mimicking normal RBC, stomatocyte and echinocyte (with six equidistant identical spicules) were simulated (with constant area $\approx 16.5\ \mu m^2$) and subsequently, the OA spectra were computed over 10-1000 MHz by evaluating the integral equation employing the Monte Carlo integration method. The OA spectrum for an equivalent disc was also calculated for comparison. The sound-speed within the source region was taken as 1639 m/s and that of the surrounding medium was chosen as 1500 m/s. The first minimum of the OA spectra for disc and echinocyte appeared almost at the same location (440 MHz) when probed from an angle, $\theta=\pi/4$ with respect to the axis of symmetry. The locations for the first minimum became 280 and 390 MHz, respectively for normal RBC and stomatocyte (for $\theta=\pi/4$). These 0A spectral features may be useful for morphological characterization of normal and deformed RBCs.

Keywords: Optoacoustics, Photoacoustics, inhomogeneous wave equation, Green's function, modified Green's function, erythrocyte, stomatocyte, echinocyte.

Introduction

Red blood cells (RBCs) under normal biophysical and biochemical condition retain biconcave shape [1, 2] maximizing surface to volume ratio, which help them to perform regular physiological activities efficiently. Some hereditary disorders make RBCs stiff and deformed posing difficulties to perform their regular functions. Stomatocytosis (ST) and echinocytosis (ET) are such disorders [1]. Efforts have been made to measure optoacoustic (OA) signals from individual cells *in vitro* [3]. Several theoretical studies have also been conducted to examine OA emission from normal and pathological cells both in the time and frequency domains [4]. It has been concluded that the OA spectral features depend upon the size and shape of the source and also on the direction of the measurement. The speed of sound inside the RBC is approximately 10% more than that of the saline water. Therefore, it acts as an acoustically inhomogeneous source. Recently, it has been shown that the OA field from an inhomogeneous source can be accurately and rapidly calculated using a modified Green's function (MGF) approach [1, 5]. In this work, we employ the MGF method to calculate the OA fields from healthy and diseased RBCs and investigate how spectral features vary with disease/shape.

Theoretical components and numerical implementation

The time-independent OA wave equation inside and outside the source can be cast as,

$$\nabla^2\psi(\mathbf{r}) + (k_f^2 + i\epsilon)\psi(\mathbf{r}) = -S(\mathbf{r}) - V(\mathbf{r})\psi(\mathbf{r}), \tag{1}$$

where ϵ is an infinitesimally small real number; $S(\mathbf{r})$ and $V(\mathbf{r})$ denote the source term and the scattering potential, respectively and defined as,

$$S(\mathbf{r}) = \begin{cases} -\frac{i\mu\beta I_o\omega}{C_p}, & \text{inside the source} \\ 0, & \text{outside the source} \end{cases} \tag{2}$$

similarly,

$$V(\mathbf{r}) = \begin{cases} k_s^2 - k_f^2 - i\epsilon, & \text{inside the source} \\ -i\epsilon, & \text{outside the source.} \end{cases} \tag{3}$$

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Here, μ, β and C_p are the optical absorption coefficient, isobaric thermal expansion coefficient and specific heat for the absorbing region, respectively; ω are I_0 are the modulation frequency and intensity of laser radiation, respectively; $k_s = \omega/v_s$ and $k_f = \omega/v_f$ are the wave numbers inside and outside the OA source, respectively with v_s and v_f being the speed of sound inside and outside the PA source. A schematic is presented in Fig. 1.

The solution to Eq. (1) can be obtained using the Green's function method. Note that free space Green's function satisfies,

$$\nabla^2 g(\mathbf{r}|\mathbf{r}_0) + k_f^2 g(\mathbf{r}|\mathbf{r}_0) = -\delta(\mathbf{r}-\mathbf{r}_0), \tag{4}$$

where δ indicates the Dirac delta function; $\mathbf{r}_0$ and $\mathbf{r}$ are the source and field points, respectively. Eq. (4) provides $g(\mathbf{r}|\mathbf{r}_0) = \frac{\exp(ik_f|\mathbf{r}-\mathbf{r}_0|)}{4\pi|\mathbf{r}-\mathbf{r}_0|}$. The solution to Eq. (1) thus becomes,

$$\psi(\mathbf{r}) = \int_{V_{vol}} g(\mathbf{r}|\mathbf{r}_0)[S(\mathbf{r}_0) + V(\mathbf{r}_0)\psi(\mathbf{r}_0)]d^3\mathbf{r}_0, \tag{5}$$

where V_{vol} is the volume of the source. Note that that the knowledge of the wave function within the source, $\psi(\mathbf{r}_0)$, is needed in order to carry out the integration, however, it is not known *a priori*. The standard approach is to iteratively expand Eq. (5) to obtain a faithful solution [4, 5]. It has been recently shown that an approximate solution can also be obtained by consiedering only the first term of Eq. (5) and using a modified Green's function, which in the far field takes the following forms [2, 5],

$$g(\mathbf{r}|\mathbf{r}_0) \approx \frac{e^{ik_f r}}{4\pi r}\, e^{-i\mathbf{k}_s\cdot\mathbf{r}_0} \quad \text{in 3D} \quad \text{and} \quad g(\mathbf{r}|\mathbf{r}_0) \approx \frac{i}{4}\sqrt{\frac{2}{\pi k_f r}}\, e^{i(k_f r-\pi/4)} e^{-i\mathbf{k}_s.r_0} \quad \text{in 2D}. \tag{6}$$

Therefore, for this Green's function, one calculates (in 2D),

$$\psi_{mgf}(\mathbf{r}) \approx -\int_{Area} \frac{i\mu\beta I_o\omega}{C_p}\frac{i}{4}\sqrt{\frac{2}{\pi k_f r}}\, e^{i(k_f r-\pi/4)} e^{-i\mathbf{k}_s.r_0} d^2\mathbf{r}_0. \tag{7}$$

The integration in Eq. (7) can be performed analytically for a solid disc (of radius a) yielding,

$$\psi_{mgf}(\mathbf{r}) = \frac{i\mu\beta I_0\omega}{4C_P}\sqrt{\frac{2}{\pi k_f}}\frac{2\pi a J_1(k_s a)}{k_s}\,\frac{e^{i(k_f r-\pi/4)}}{\sqrt{r}}. \tag{8}$$

The OA spectra for normal and pathological RBCs were computed by numerically computing Eq. (7). For simplicity, we considered 2D shapes only. The same spectra for the Green's function (GF) method were also generated comparison. The Monte Carlo integration was employed. Table 1 details various morphological parameters (D=diameter, t=dimple thickness, h=maximum thickness, d=diameter of the circle passing through the highest thickness for RBC; N_s=number of spicules, a_{sp}=radius of the inner disc, h_s and b_s height and base of half ellipse, respectively for ET). The area remained almost constant for all the shapes. These shapes are displayed in Fig. 2(a)-(d).

Numerical results

The OA spectra for normal and pathological RBCs are presented in Fig. 2(e)-(l). Each figure contains spectra generated by the GF and MGF methods. It is clear that OA spectrum in general exhibits an oscillatory pattern.

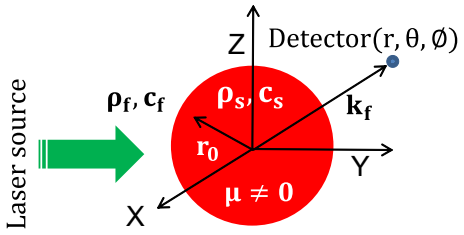


Figure 1: Source-detector arrangement.

PA Sources	Morphological parameters (μm)	Area in μm^2
Circular disc	r=2.2827	16.33
Normal RBC	D=7.65, t=1.44, d=5.355, h=2.84	16.50
ST	D=5.25, t=1.40, d=3.675, h=2.84	16.41
EC	N_s=6, a_{sp}=1.96, h_s=1.0, b_s=0.5	16.65

Table 1: Morphological specifications for different OA sources.

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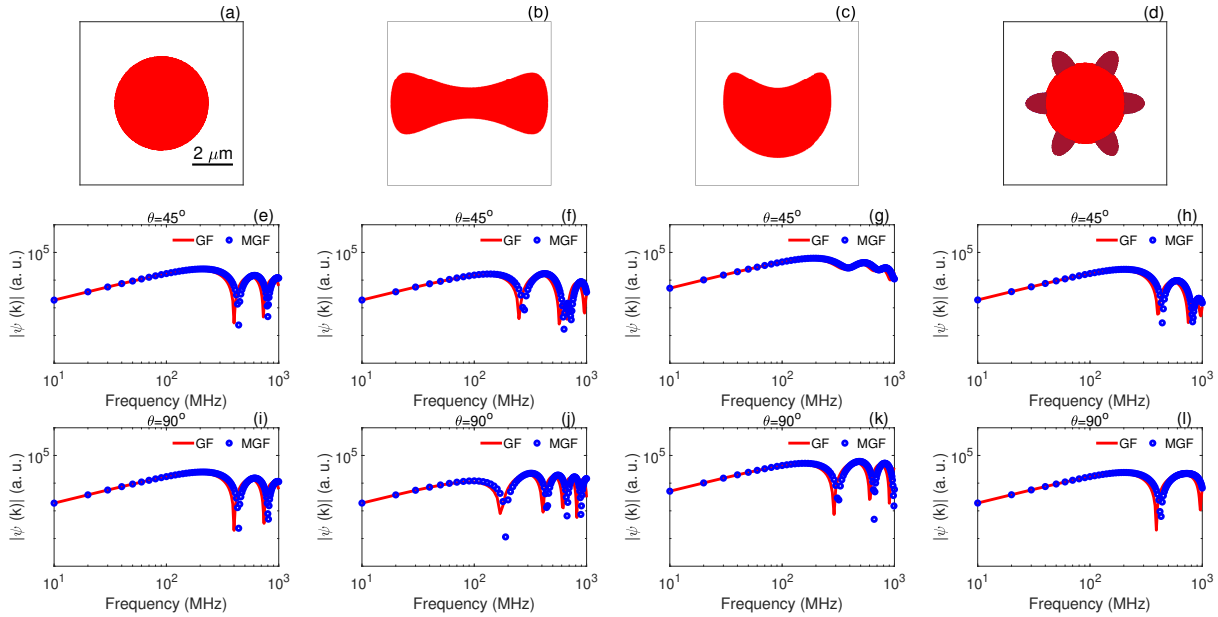


Figure 2: (a)-(d) Represent a circular disc, normal RBC, ST and ET in 2D, respectively. (e)-(l) Illustration of the OA spectra for these sources (for second row $\theta = \pi/4$ and for the third row $\theta = \pi/2$).

The spectra for disc and ET look similar [the first minimum occurs at 400 MHz; see (e) and (h)]. The first minimum for RBC appears much earlier (at 250 MHz) than that of the disc when probed from $\theta = 45^o$ [see (e) and (f)]. The same arises even earlier for normal RBC [at around 170 MHz; see (j)] when computed from $\theta = 90^o$. The dips for the ST shape are not so prominent for $\theta = 45^o$ compared to $\theta = 90^o$.

Discussion and conclusions

In this work, the MGF approach has been used to calculate the OA spectra for normal and pathological RBCs. In this method, the mismatch in speed of sound is incorporated within the Green's function which results in better estimation of OA spectrum compared to the GF method [2]. It is worthy to mention that the locations of minima for each case in the MGF scheme appear little later than those of the GF protocol. It is expected because as the speed of sound inside the source is increased then the OA pulse width will decrease as a result of that spectral minimum will appear at a higher frequency. This work may help to design an OA spectral analysis protocol for morphological characterization of cells.

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